Economic Growth and Human Well-being: Visualizing Paths of Progress Across Countries

STATS 401 · Final Project

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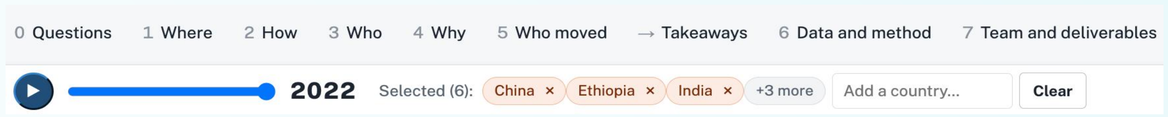
INTRODUCTION AND RESEARCH QUESTIONS

Does a country's income tell us how well its people live? GDP is a common yardstick of progress, but health, schooling, services, work and clean air can follow different paths.

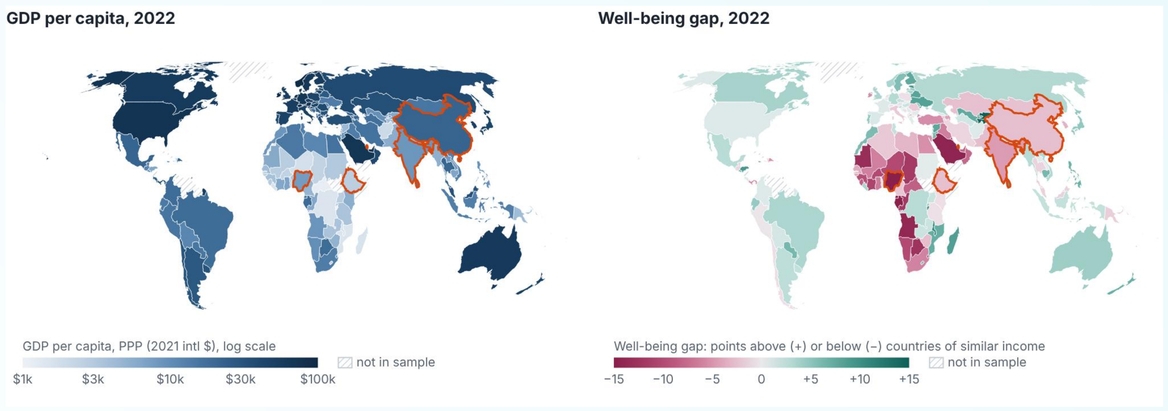
Research Question:

1. **How does prosperity vary across countries and over time?**
2. **How have countries followed different development paths?**
3. **How strongly is income associated with well-being, and who departs from it?**
4. **Which dimensions explain the departures?**
5. **Which countries moved to a higher stage of development?**

Coordinated exploration. One shared year, country selection and dimension weights link all five D3 views. Hover, click, brush and the guided tour let readers follow a question across views.

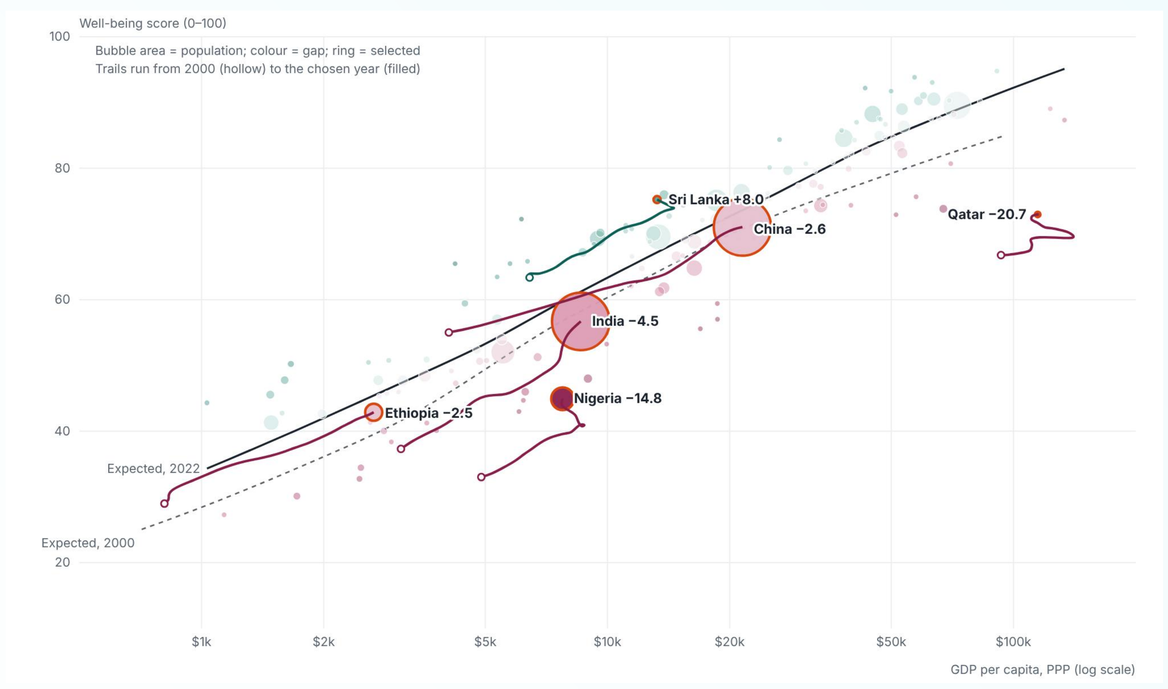


1 WHERE? PAIRED CHOROPLETH MAPS



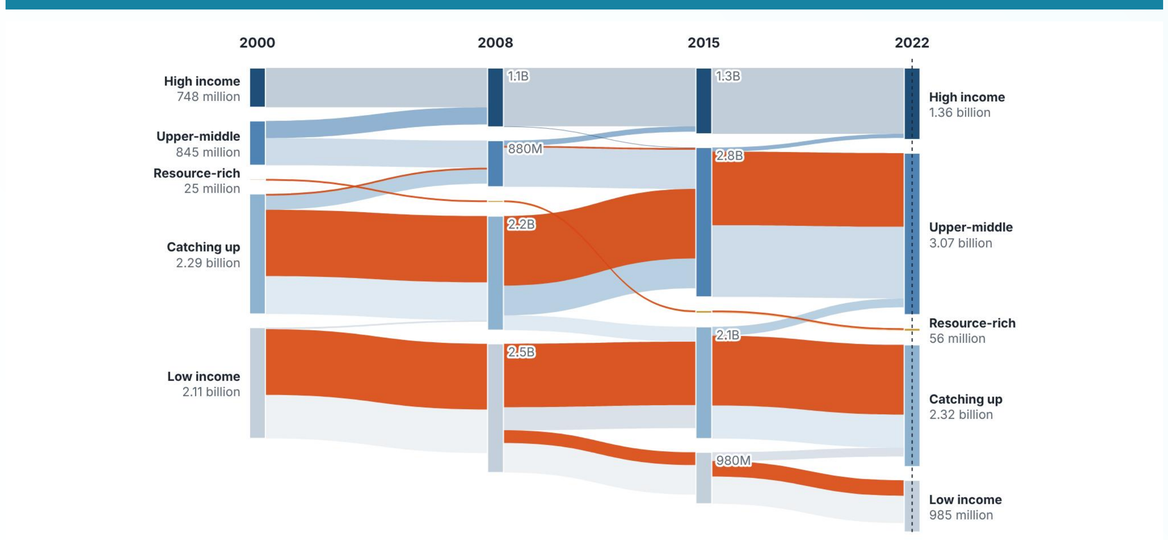
Finding: Richest regions do not consistently outperform their income. Gulf states and Central African oil producers often fall short, while parts of the former Soviet Union and East Africa outperform.
Interaction: Linked zoom and pan, measure switching and country selection.

2 HOW? PATHS OF PROGRESS



Finding: At $2,000 per person, expected well-being rose from 36 to 42. China moved from above to below expectation, while Sri Lanka remained above.
Interaction: Animate years, switch score/gap, brush groups and follow selected trails.

5 WHO MOVED? DEVELOPMENT PROFILES



Finding: 33/42 "catching up" countries reached upper-middle, and 23/32 upper-middle countries reached the high profile. The lowest-profile population share fell from 35% to 13%.
Interaction: Switch country/population weighting and select ribbons or blocks. Orange overlays mark selected countries.

DATA AND METHODS

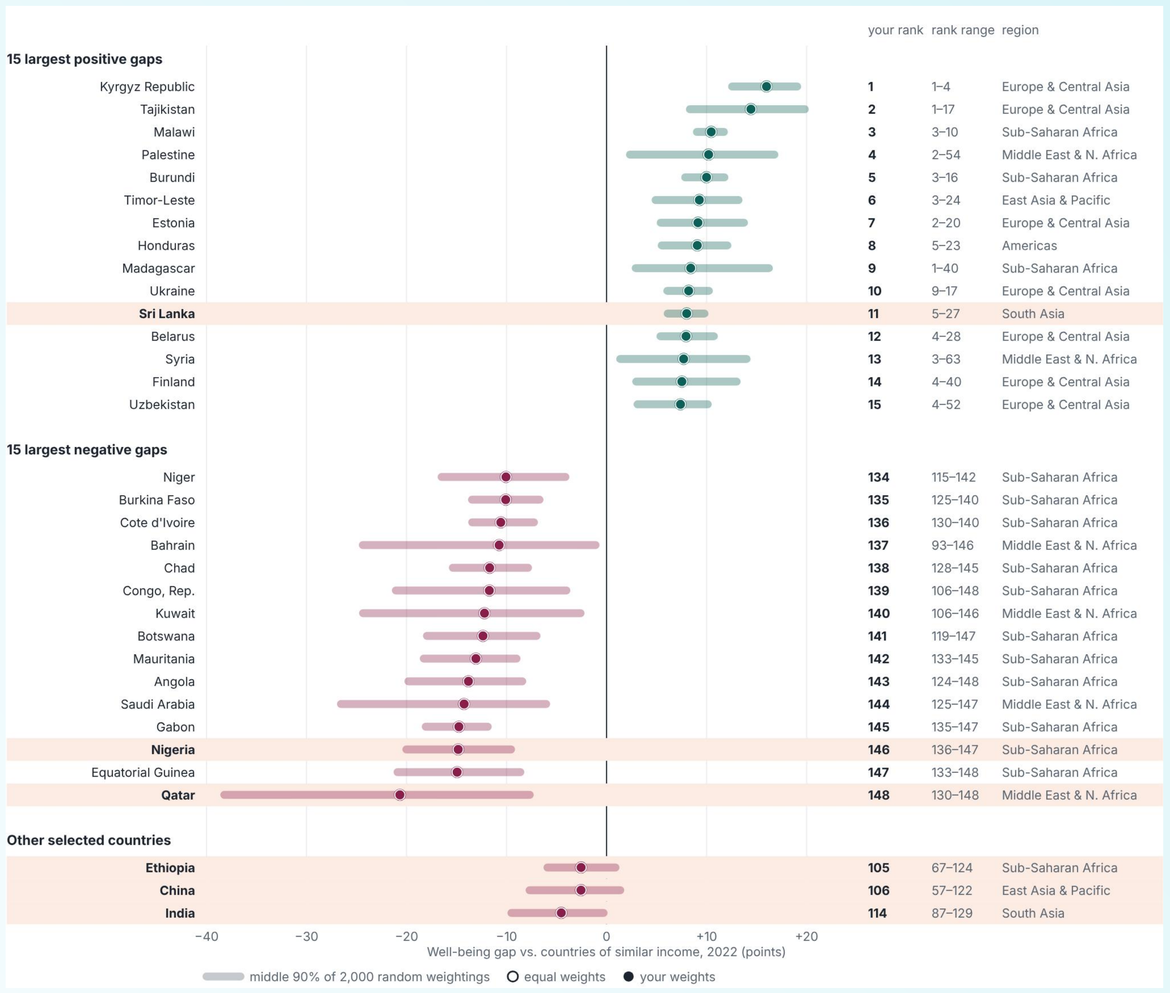
Data. World Bank WDI, UNDP HDR, World Happiness Report and Natural Earth. A balanced panel of 148 countries, 2000–2022 (3,404 country-years), with interpolation and edge fills.

Measure. Ten indicators form five dimensions and an equal-weight score. Well-being gap = score − yearly LOWESS expectation at log GDP per capita.

Analysis. 2,000 Dirichlet weightings test sensitivity. Pooled k-means (k = 5) defines development profiles at 2000, 2008, 2015 and 2022.

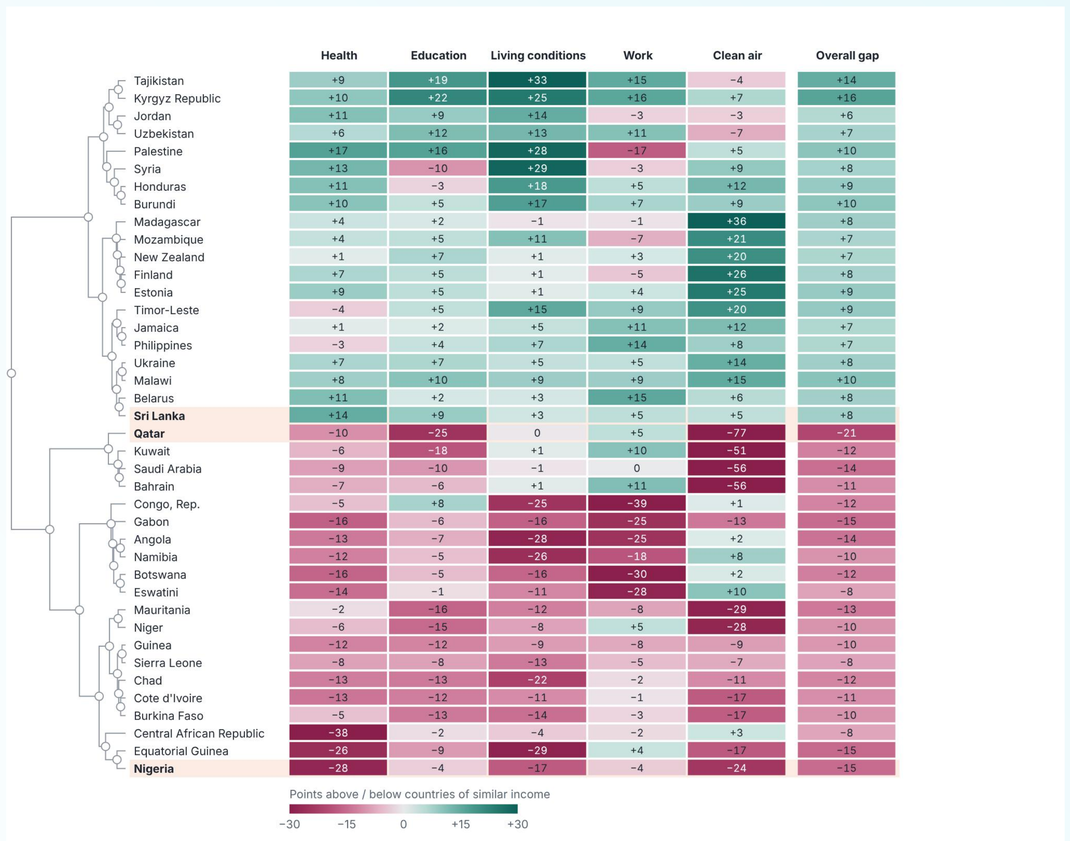
Validation. Life satisfaction correlates with the score (r = 0.78) and, after income adjustment, with the gap (r = 0.28), across 2,053 country-years.

3 WHO? RANKINGS AND WEIGHT SENSITIVITY (2022)



Finding: Kyrgyz Republic, Malawi and Burundi remain near the top across weights. Palestine and Madagascar vary more. Qatar is last under equal weights, with a wide sensitivity range.
Interaction: Reweight five dimensions, re-rank and filter regions. Bars show the middle 90% of random weightings.

4 WHY? CLUSTERED DIMENSION GAPS (2022)



Finding: Gulf shortfalls concentrate in clean air and education; Sub-Saharan resource exporters fall short in health, living conditions and work; Central Asian gains reflect schooling and services.
Interaction: Sort dimensions, switch to selected countries or click a Ward-cluster branch to select the group.

Conclusion. Income is a strong but imperfect guide to well-being. Coordinated views reveal uneven development paths and distinct shortfalls. Limits include natural desert dust, GDP collapses and partly projected schooling.

Sources: World Bank WDI · UNDP HDR 2023/24 · World Happiness Report · Natural Earth

Explore the coordinated views and reproducible analysis: yyiffann.github.io/stats401-final-project/